

RAI-818: INTELLIGENT **MOBILE ROBOTICS**

- **Grid Based Approach**
- **Navigation Functions**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

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Potential Fields

- Potential Fields
 - Grid Based Approach

Grid Based Approach

- Uses a discrete version of the configuration space
- *Grid* is a two dimensional array of square elements called *pixels*
- The *Brushfire* algorithm is one way to use the grid based approach
 - A *pixel* has a value of zero when it is completely free
 - A *pixel* has a value of one if it is completely or partially by an obstacle
 - Planner has a choice for determining grid connectivity, i.e.,

<i>n1</i>	<i>n2</i>	<i>n3</i>
<i>n4</i>	<i>n5</i>	<i>n6</i>
<i>n7</i>	<i>n8</i>	<i>n9</i>

4 Point Connectivity –
Manhattan distance
Only North, South, East and West are connected

<i>n1</i>	<i>n2</i>	<i>n3</i>
<i>n4</i>	<i>n5</i>	<i>n6</i>
<i>n7</i>	<i>n8</i>	<i>n9</i>

8 Point Connectivity



Defining futures

Potential Fields



- Potential Fields
 - Grid Based Approach

Grid Based Approach : Brushfire Algorithm

- In the first step free space *pixels* are given a value of 0 and obstacles have a value of 1
- Next all 0 valued *pixels* neighboring 1 value *pixels* are labeled with 2 (4 or 8 point connectivity) is used to determine adjacency
- Next all 0 valued *pixels* neighboring 2 value *pixels* are labeled with 3
- Repeat this procedure until all 0 valued *pixels* adjacent to an i are labeled as $i+1$
- Stop when all *pixels* have an assigned value
- The final result is a distance map d where each cell holds the minimum distance to an obstacle.
- The gradient of distance is easily determined by finding a neighboring cell with the lowest *pixel* value



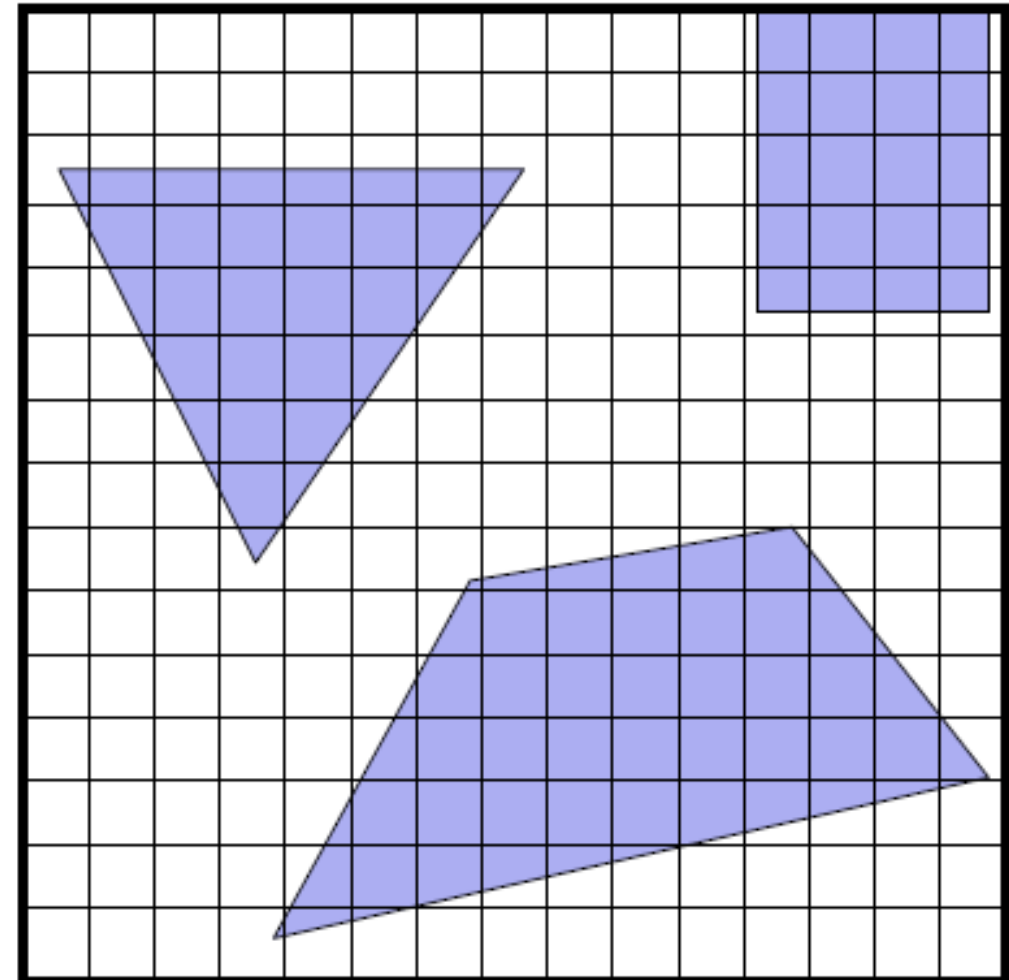
Defining futures

Potential Fields



- Potential Fields
 - Grid Based Approach

Grid Based Approach : Brushfire - Example



Defining futures

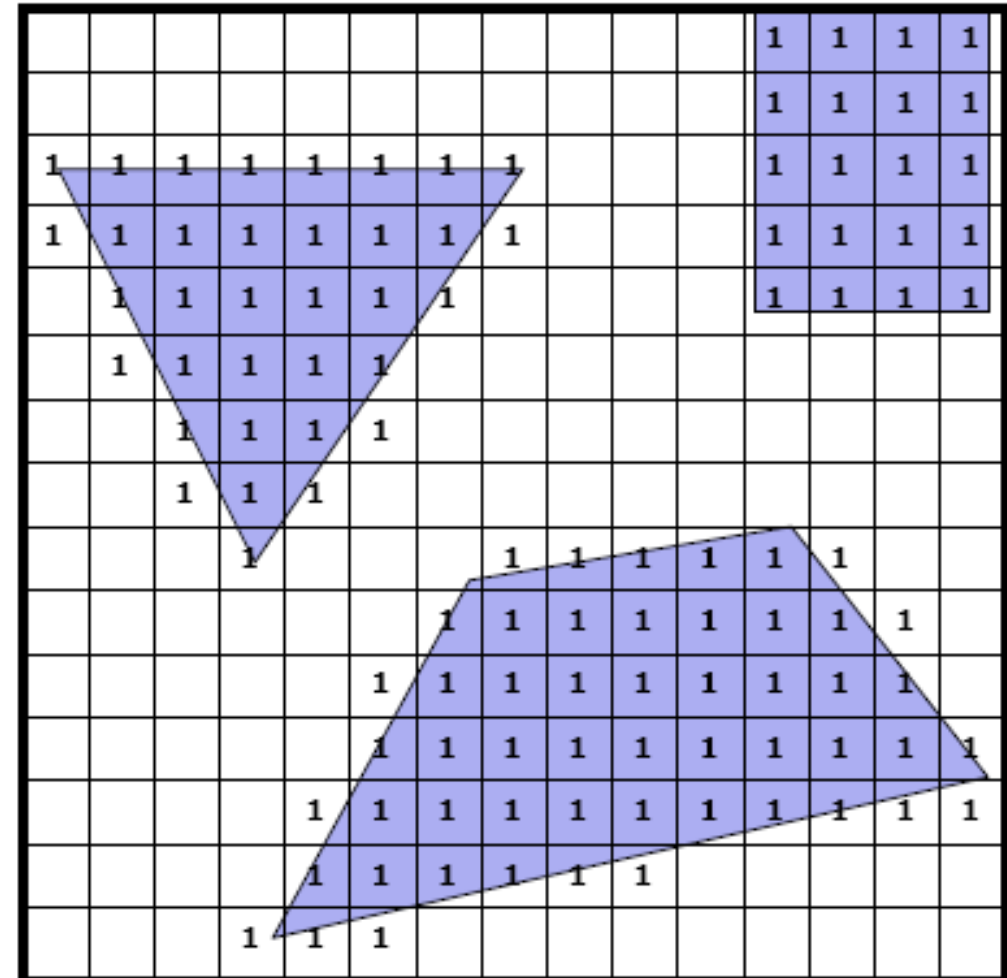
Potential Fields



- Potential Fields
 - Grid Based Approach

Grid Based Approach : Brushfire - Example

- Start with an empty grid with *obstacles* equal to 1



Defining futures

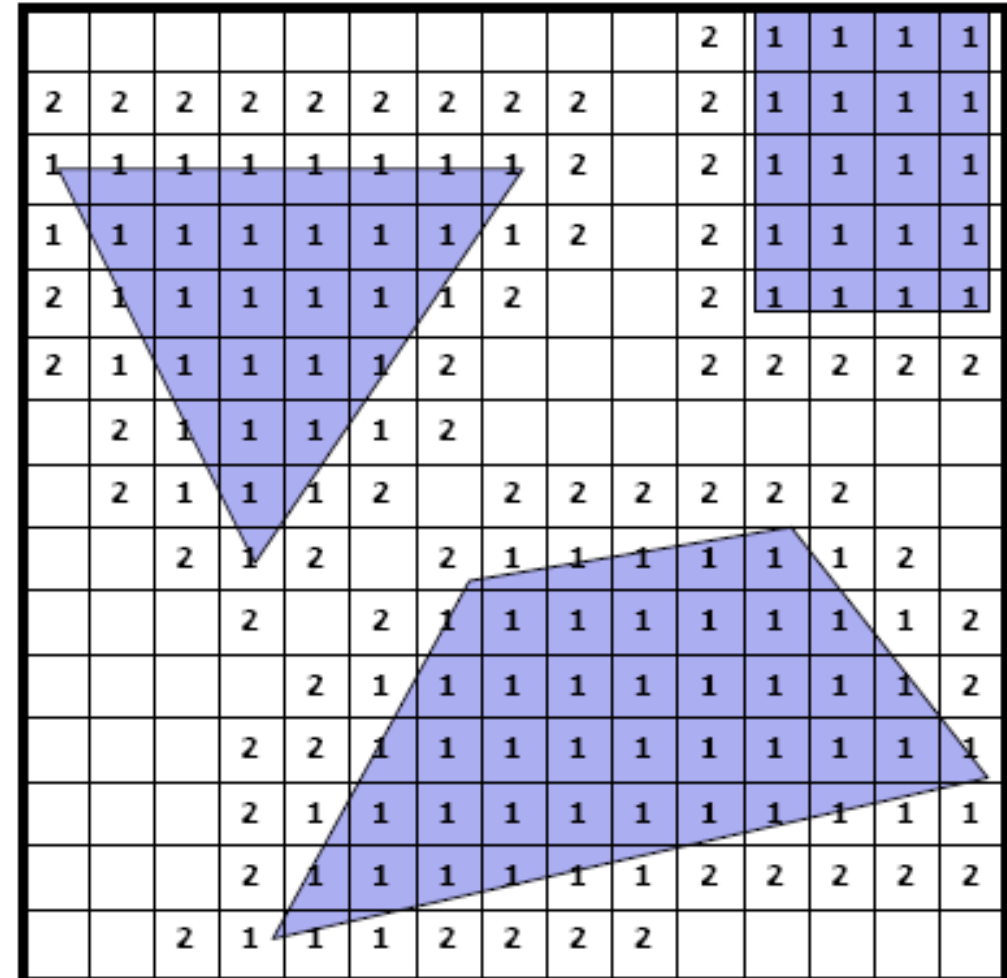
Potential Fields



- Potential Fields
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Grid Based Approach : Brushfire - Example

- Start with an empty grid with *obstacles* equal to 1
- *Expand* cells i to $i+1$



Defining futures

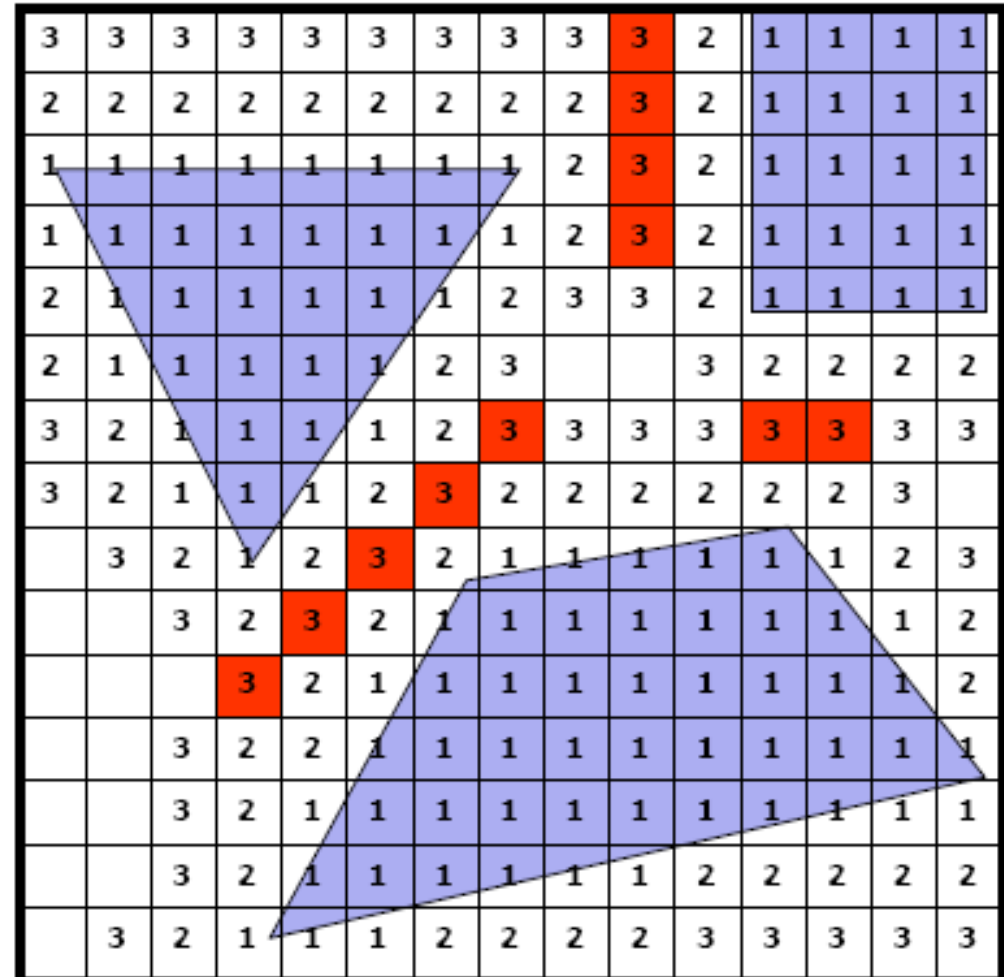
Potential Fields



- Potential Fields
 - Grid Based Approach

Grid Based Approach : Brushfire - Example

- Start with an empty grid with *obstacles* equal to 1
- *Expand* cells i to $i+1$
- *Repeat*



Defining futures

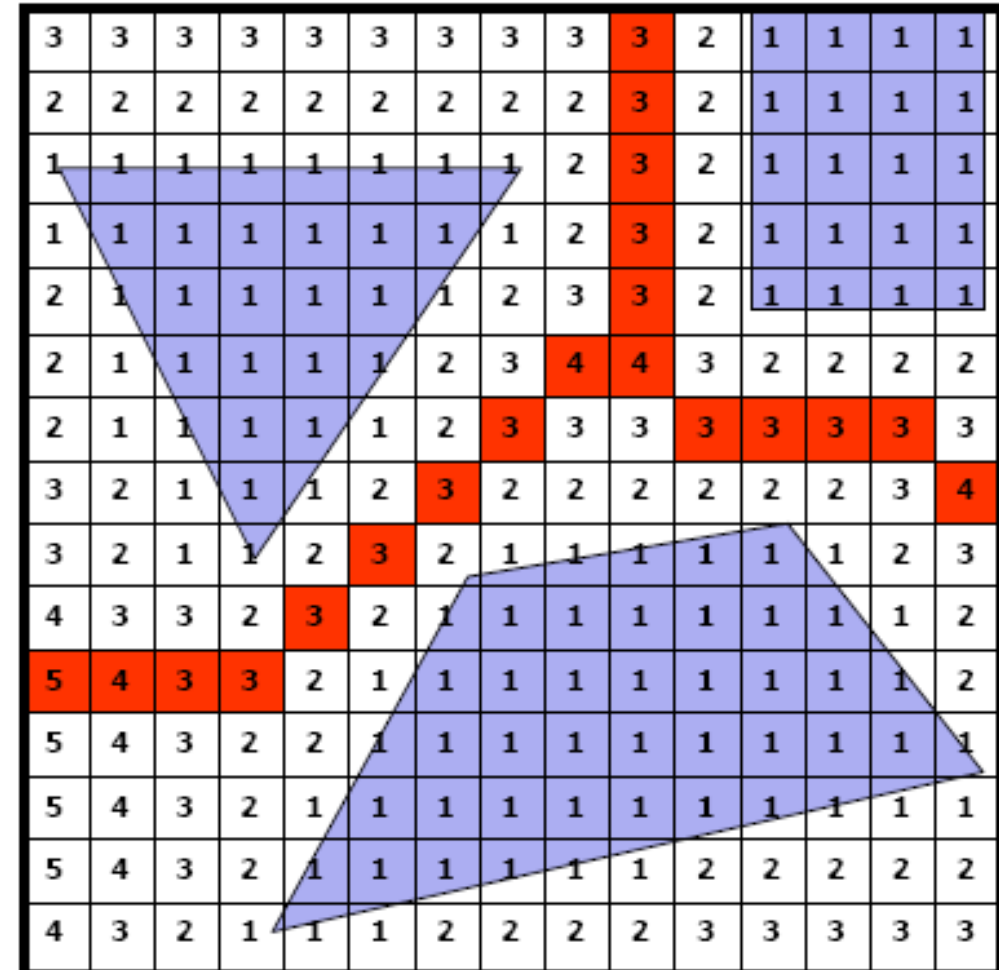
Potential Fields



- Potential Fields
 - Grid Based Approach

Grid Based Approach : Brushfire - Example

- Start with an empty grid with *obstacles* equal to 1
- *Expand* cells i to $i+1$
- *Repeat*
- *Repeat Again*



Defining futures

Potential Fields



- Potential Fields
 - Grid Based Approach

Grid Based Approach : Brushfire Algorithm Formally

Input

A grid of pixel where free space pixels have a value of 0 and obstacle pixels have a value of 1

Output

A grid of pixels whose values measure distance to the nearest obstacle

- 1: Label all the *obstacle* pixels as 1
- 2: create queue *L* of *pixels* on the boundary of all obstacles
- 3: **while** *L* $\neq$ 0 **do**
- 4: *pop* the *top* element *t* of *L*
- 5: **if** $d(t) = 0$
- 6: set $d(t)$ to $1 + \min_{t' \in N(t), d(t) \neq 0} d(t')$
- 7: add all $t' \in N(t)$, with $d(t) = 0$ to *L*
- 8: **end while**



Defining futures

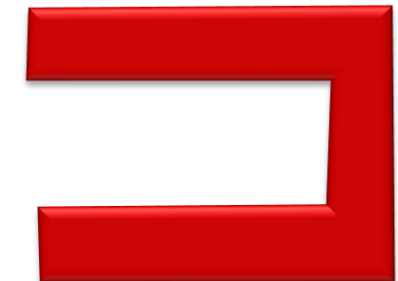


Potential Fields

- **Potential Fields**
 - Grid Based Approach
 - **Local Minima**

Local Minimum Problem

- How do we know that we have only a single (global) minimum?
 - For appropriate choice of $\alpha(i)$ the *gradient descent algorithm* will *always converge to a minimum* in the field
 - But there is *no guarantee* that this will be the *global minimum*
 - This means that there is *no guarantee* that the *gradient descent* will *find a path to q_{goal}*
 - In the *figure* shown the *robot is initially attracted to the goal* as it *approaches* the (*horseshoe shaped*) obstacle.
 - The *goal attracts the robot* while the *bottom arm deflects the robot upward* until the *top arm begins to deflect downward*
 - Robot *reaches point* where *effect of obstacle's base balances attraction of the goal* - $\nabla U(q^*) = 0$ and q^* is not goal
 - This makes the *potential field planner incomplete*



q_{goal}



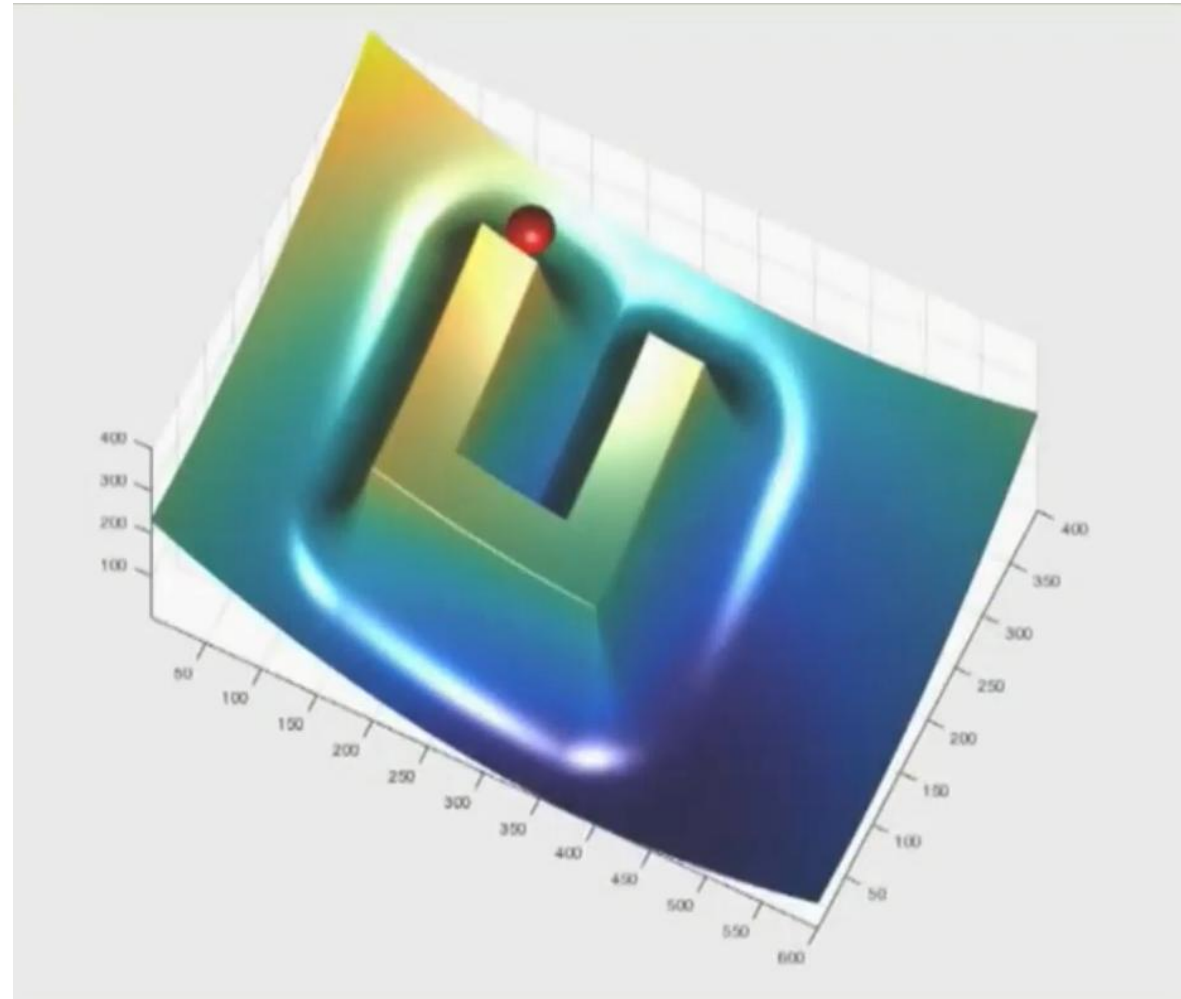
Defining futures

Potential Fields



- **Potential Fields**
 - Grid Based Approach
 - **Local Minima**

Local Minimum Problem



Defining futures

Potential Fields



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 - Local Minima
 - **Navigation Functions**

Navigation Functions

- **We have two choices:**
 - **not guaranteed to be a global minimum: do something other than gradient descent (what?)**
Randomized Path Planner (RPP)
 - **make sure only one global minimum (e.g., use a navigation function)**
- **Navigation functions are a class of potential functions which fulfil a number of constraints**
 - **Goals are minima**
 - **Obstacles are maxima**
 - **There are no local extrema besides goals and obstacles**
- **Navigation functions form complete and correct path planners**



Defining futures



Potential Fields

- **Potential Fields**
 - Grid Based Approach
 - Local Minima
 - **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – Part I

Simplest solution to the local minima problem – only for grid based approaches

*Label the **Goal pixel 2**, the **obstacle pixel 1** and the **free pixels 0***

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
3	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15



Defining futures



Potential Fields

- **Potential Fields**
 - Grid Based Approach
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 - **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – Part I

- Starting with the goal, set all adjacent cells with **0** to the current cell + **1**
 - **4Point Connectivity** or **8-Point Connectivity**?
 - **Your Choice.** We'll use **8-Point Connectivity** in our example

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
3	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	0	3	3	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	2	
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15



Defining futures



Potential Fields

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Navigation Functions – Wave Front Planner – Part II

- Now repeat with the modified cells
 - This will be repeated until no **0**'s are adjacent to cells with values ≥ 2
 - **0**'s will only remain when regions are **unreachable**

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
3	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	0	0	0	4	4	4
1	0	0	0	0	0	0	0	0	0	0	0	0	0	4	3	3
0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	3	2
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15



Defining futures



Potential Fields

- **Potential Fields**
 - Grid Based Approach
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 - **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – Part III

- Repeat Again

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0
3	0	0	0	0	1	1	1	1	1	1	1	1	5	5	5	5
2	0	0	0	0	0	0	0	0	0	0	0	0	5	4	4	4
1	0	0	0	0	0	0	0	0	0	0	0	0	5	4	3	3
0	0	0	0	0	0	0	0	0	0	0	0	0	5	4	3	2
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15



Defining futures



Potential Fields

- **Potential Fields**
 - Grid Based Approach
 - Local Minima
 - **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – Part IV

- **And Again**

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	1	1	1	1	1	1	1	1	6	6	6	6
3	0	0	0	0	1	1	1	1	1	1	1	1	5	5	5	5
2	0	0	0	0	0	0	0	0	0	0	0	6	5	4	4	4
1	0	0	0	0	0	0	0	0	0	0	0	6	5	4	3	3
0	0	0	0	0	0	0	0	0	0	0	0	6	5	4	3	2
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15



Defining futures



Potential Fields

- **Potential Fields**
 - Grid Based Approach
 - Local Minima
 - **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – Part V

- **And Again Until...**

7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	7	7	7	7	7	7
4	0	0	0	0	1	1	1	1	1	1	1	1	6	6	6	6
3	0	0	0	0	1	1	1	1	1	1	1	1	5	5	5	5
2	0	0	0	0	0	0	0	0	0	0	7	6	5	4	4	4
1	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	3
0	0	0	0	0	0	0	0	0	0	0	7	6	5	4	3	2
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15





Potential Fields

- **Potential Fields**
 - Grid Based Approach
 - Local Minima
 - **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – Done

- You're done
 - Remember, 0's should only remain if unreachable regions exist

7	18	17	16	15	14	13	12	11	10	9	9	9	9	9	9
6	17	17	16	15	14	13	12	11	10	9	8	8	8	8	8
5	17	16	16	15	14	13	12	11	10	9	8	7	7	7	7
4	17	16	15	15	1	1	1	1	1	1	1	1	6	6	6
3	17	16	15	14	1	1	1	1	1	1	1	1	5	5	5
2	17	16	15	14	13	12	11	10	9	8	7	6	5	4	4
1	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3
0	17	16	15	14	13	12	11	10	9	8	7	6	5	4	2
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14



Potential Fields

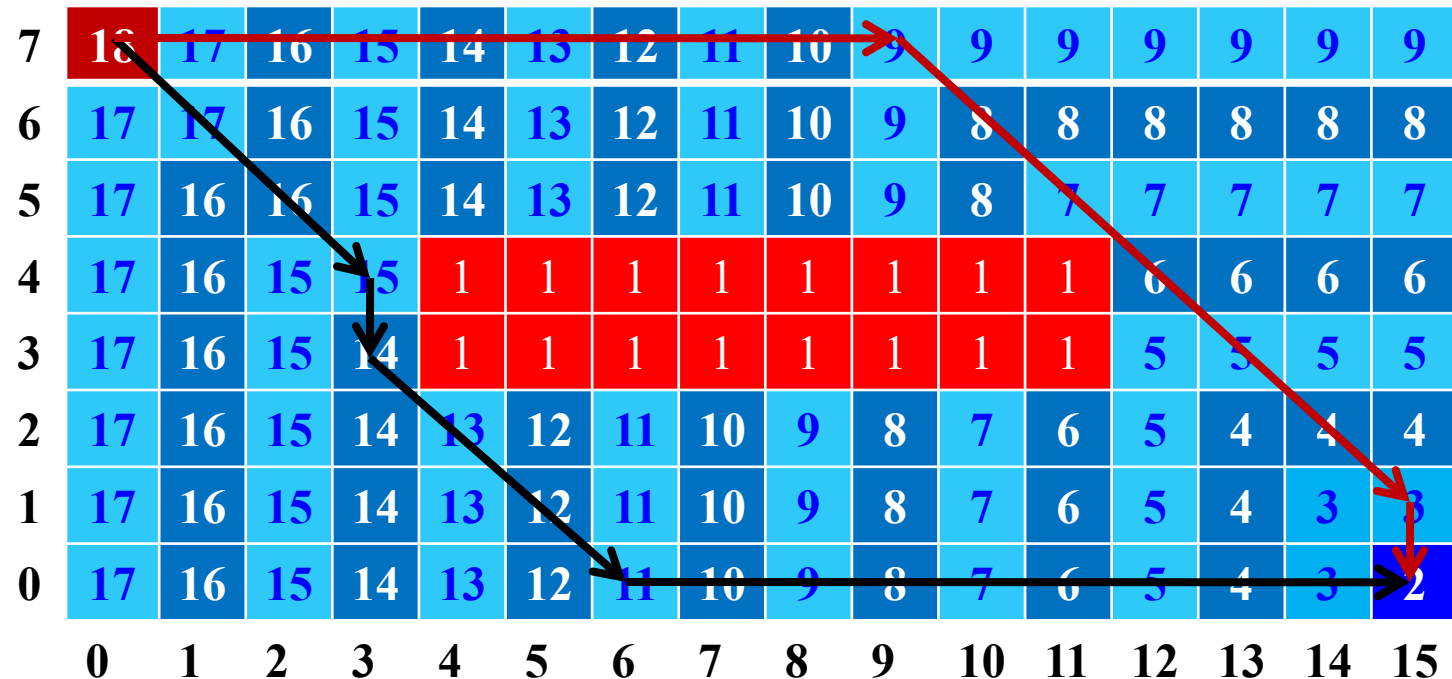


- **Potential Fields**
 - Grid Based Approach
 - Local Minima
- **Navigation Functions**
 - Wave Front Planner

Navigation Functions – Wave Front Planner – What Now?

- To find the shortest path, according to your metric simply always move toward a cell with a lower number
 - The numbers generated by the Wavefront planner are roughly proportional to their distance from the goal

Two possible shortest paths



Defining futures



Potential Fields

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Wave-Front Algorithm - Formally

Input

A grid of pixel where free space pixels have a value of 0, obstacle pixels have a value of 1 and the goal pixel has a value of 2

Output

A grid of pixels whose values measure distance to the nearest obstacle

- 1: *Label* the *goal pixel* as 2
- 2: **Label** all the *obstacle pixels* as 1
- 3: *create queue L* of *pixels* on the *boundary* of all *obstacles* with value 0
- 4: **while** $L \neq 0$ **do**
- 5: *pop* the *top element t* of L
- 6: *set* $d(t)$ to $1 + \min_{t' \in N(t), d(t) > 1} d(t')$
- 7: *add* all $t' \in N(t)$, with $d(t) = 0$ to L
- 8: **end while**



Potential Fields



- **Potential Fields**
 - Grid Based Approach
 - Local Minima
 - **Navigation Functions**
 - Wave Front Planner
 - **Navigation Potential Function**

Navigation Potential Function

- So far we have seen that
 - **Bug Algorithms** are **complete sensor based planners** that work in unknown spaces but are limited to **planar configuration spaces**
 - **Attractive/repulsive potential field** approach applies to a **general class of configuration spaces** but suffers from **local minima** problems and hence it is **not complete**
- The **Wave-front planner** address the **local minima** problem but requires time and **storage exponential** in the **dimension of the space**
- So we introduce a **new potential** that is a **function of the distance** to the obstacles, has **only one minimum** and **applies to a limited class of configuration spaces** with dimensions 2, 3 and more
- Such potential functions are called *navigation functions*



Defining futures



Potential Fields

- **Potential Fields**
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 - Wave Front Planner
 - Navigation Potential Function

Navigation Potential Function

- A function $\phi: Q_{free} \rightarrow [0, 1]$ is called a *navigation function* if it
 - is *smooth* (or at least C^k for $k \geq 2$)
 - has a *unique minimum* at
 - is *uniformly maximal* on the *boundary* of free space
 - is *Morse*
- A function is *Morse* if every critical point is isolated.
(a point where the gradient is zero)
- The Navigation Function approach represents obstacles as
$$QO_i = \{q | \beta_i(q) \leq 0\}$$
 - $\beta_i(q)$ is *-’ve* in the *interior* of QO_i
 - $\beta_i(q)$ is **0** at the *boundary* of QO_i
 - $\beta_i(q)$ is *+’ve* in the *exterior* of QO_i



Defining futures

Potential Fields



- **Potential Fields**
 - Grid Based Approach
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 - Wave Front Planner
 - Navigation Potential Function
 - **Sphere World**

Sphere World

- Suppose that the world is a *sphere* of radius r_o centered at q_o containing n obstacles of *radius* r_i centered at q_i , $i = 1 .. n$
- The obstacle distance functions are easy to define as
 - $\beta_o(q) = -d^2(q, q_o) + r_o^2$
 - $\beta_i(q) = d^2(q, q_i) - r_i^2$ $\beta_i(q)$ increases continuously as q moves away from the obstacle

Repulsive Function

- Instead of considering distance to the closest obstacle or distance to each obstacle we consider

$$\beta(q) = \prod_{i=0}^n \beta_i(q)$$

Attractive Function

- Defined as a *power function* of distance to goal or $\gamma_\kappa(q) = (d(q, q_{goal}))^{2\kappa}$
 - note this will be **0** at the *goal*, and *increasing* as we move away
 - κ controls the *rate of growth*



Defining futures



Potential Fields

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 - Navigation Potential Function
 - **Sphere World**

Sphere World

- Consider now the function $\frac{\gamma_{\kappa}}{\beta}(q)$
The function $\frac{\gamma_{\kappa}}{\beta}(q)$
 - is only **0** at the *goal*
 - goes to ∞ at the *boundary* of any *obstacle*
 - for a large enough value of κ , has a *unique minimum*
- Effects of increasing κ
 - $\frac{\partial \gamma_{\kappa}}{\partial(q)}$ dominates $\frac{\partial \beta}{\partial(q)}$ making the gradient point towards the goal
 - Causes *critical points* to gravitate towards the obstacles as the range of the *repulsive influence* of obstacles becomes *small* relative to the *overwhelming influence* of the *attractive field*
 - There cannot be any local minimum for large κ other than the *goal* (the *unique minimum* is the *goal*)





Potential Fields

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Sphere World

- **One Problem** : $\frac{\gamma_\kappa}{\beta}(q)$ has a unique minimum but it can have arbitrarily large values

- **Solution** : Introduce an analytical switch $\sigma_\lambda(x) = \frac{x}{\lambda + x}, \lambda > 0$
 - Since $\sigma_\lambda(x)$ is 0 at $x = 0$ and converges to 1 as x approaches ∞ it can be used to bound the value of $\frac{\gamma_\kappa}{\beta}(q)$ i.e.,

$$s(q, \lambda) = \left(\sigma_\lambda \circ \frac{\gamma_\kappa}{\beta} \right) (q) = \left(\frac{\gamma_\kappa}{\lambda\beta + \gamma_\kappa} \right) (q)$$

the function $s(q, \lambda)$ has a value 0 at the *goal*, 1 at the *boundary* of any obstacle and *varies continuously* in the *free space*

- **Another Problem** : $s(q, \lambda)$ may not be Morse
- **Solution** : Introduce a sharpening function $\xi_\kappa(x) = x^{1/\kappa}$
- For $\lambda = 1$ the resulting navigation function on a sphere world is given by

$$\phi(q) = \left(\xi_\kappa \circ \sigma_\lambda \circ \frac{\gamma_\kappa}{\beta} \right) (q) = \frac{d^2(q, q_{goal})}{[(d^2(q, q_{goal}))^{2\kappa} + \beta(q)]^{1/\kappa}}$$

This is guaranteed to have a single minimum at q_{goal} for a sufficiently large κ





Potential Fields

- **Potential Fields**

- Grid Based Approach
- Local Minima
- Navigation Functions
- **Potential Functions for Rigid Bodies**

Potential Functions for Rigid Bodies

- Our goal is to construct a potential function that repels the robot from obstacles and attracts it towards a global minimum that is the goal
- It is easy to do this in the configuration space since the robot is a point
- In the workspace the robot has a finite area and a volume in 3D . So evaluating the effect of a potential field on a robot would involve computing an integral over the area/volume which can be very complex
- Alternative is to select a subset of points i.e. *control points* and define a potential for each of these points
- Evaluating the effect of the potential field on a single *control points* is no different from the previous evaluations
- Then we use the relation that we just developed to convert individual workspace forces into configuration space forces
- Finally we add the forces to get the final total configuration space force



Defining futures

Potential Fields



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Potential Functions for Rigid Bodies

- We need to pick *control points* $\{r_j\}$ on the robot. Min # *control points* depends on the *d.o.f.* of the robot and as before it is the number of points required to *pin down* the robot in its configuration space
- For a rigid body robot in the plane we can *pin down* the robot using just two points

- For each r_j the attractive potential is

$$U_{att,j}(q) = \begin{cases} 1/2\zeta_j d^2(r_j(q), r_j(q_{goal})), & d(r_j(q), r_j(q_{goal})) \leq d_{goal}^* \\ d_{goal}^* \zeta_j d(r_j(q), r_j(q_{goal})) - 1/2\zeta_j d_{goal}^*, & d(r_j(q), r_j(q_{goal})) > d_{goal}^* \end{cases}$$

- With this potential function the workspace force for attractive control point r_j is defined by

$$\nabla U_{att,j}(q) = \begin{cases} \zeta_j(r_j(q) - r_j(q_{goal})), & d(r_j(q), r_j(q_{goal})) \leq d_{goal}^* \\ \frac{d_{goal}^* \zeta_j(r_j(q) - r_j(q_{goal}))}{d(r_j(q), r_j(q_{goal}))}, & d(r_j(q), r_j(q_{goal})) > d_{goal}^* \end{cases}$$



Defining futures



Potential Fields

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Potential Functions for Rigid Bodies

- For the workspace repulsive potential fields we use the same control points r_j and define the repulsive potential for r_j as

$$U_{rep,j}(q) = \begin{cases} \frac{1}{2} \eta_j \left(\frac{1}{d_i(r_j(q))} - \frac{1}{Q_i^*} \right)^2, & d_i(r_j(q)) \leq Q_i^* \\ 0, & d_i(r_j(q)) > Q_i^* \end{cases}$$

where $d_i(r_j(q))$ is the shortest distance between the *control points* r_j and the obstacle WO_i and Q^* is the workspace distance of influence for the obstacles

- The gradient of each repulsive function corresponds to the workspace force

$$\nabla U_{rep,j}(q) = \begin{cases} \eta_j \left(\frac{1}{Q_i^*} - \frac{1}{d_i(r_j(q))} \right) \frac{1}{d_i^2(r_j(q))} \nabla d_i(r_j(q)), & d_i(r_j(q)) \leq Q_i^* \\ 0, & d_i(r_j(q)) > Q_i^* \end{cases}$$



Defining futures



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Potential Functions for Rigid Bodies

- Often vertices are selected as repulsive control points
- **Problem:** Selection of repulsive control points does not guarantee that no collisions will occur (as an example see the figure)
 - Here repulsive control points r_1 and r_2 are far from the obstacle WO_i
 - So repulsive influence may not be large enough to prevent collision
- **Solution:** Define a floating control point r_{float}
 - Point on the boundary of the robot closest to the obstacle
 - Choice of r_{float} depends on q

Total configuration space force acting on a robot is the sum of the configuration space forces that result from all attractive and repulsive control points

$$\begin{aligned} u(q) &= \sum_i u_{att\ j} + \sum_{ij} u_{rep\ j} \\ &= \sum_j J_j^T(q) \nabla u_{att\ j}(q) + \sum_i \sum_j J_j^T(q) \nabla u_{rep\ j} \end{aligned}$$



Defining futures

Potential Fields



- **Potential Fields**
 - Grid Based Approach
 - Local Minima
 - Navigation Functions
 - **Potential Functions for Rigid Bodies**

Potential Functions for Rigid Bodies

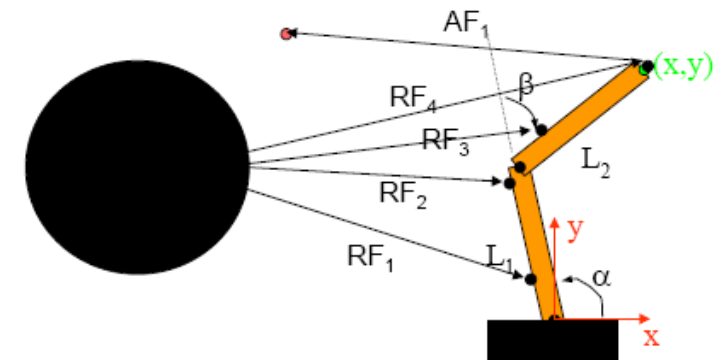
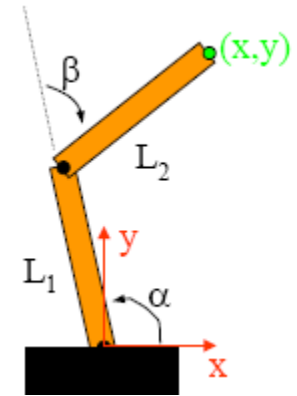
- **Example :** Our two-link Manipulator

$$J(q) = \begin{bmatrix} -L_1 \sin \alpha - L_2 \sin(\alpha + \beta) & -L_2 \sin(\alpha + \beta) \\ L_1 \cos \alpha + L_2 \cos(\alpha + \beta) & L_2 \cos(\alpha + \beta) \end{bmatrix}$$

- Suppose $q_{goal} = (0,0)^t$ then $f_w = (x,y)$

$$f_q = \begin{bmatrix} x[-L_1 \sin \alpha - L_2 \sin(\alpha + \beta)] + y[L_1 \cos \alpha + L_2 \cos(\alpha + \beta)] \\ x[-L_2 \sin(\alpha + \beta)] + y[L_2 \cos(\alpha + \beta)] \end{bmatrix}$$

- **General Procedure:**
 - Pick several points on the manipulator
 - Compute attractive and repulsive potentials for each
 - Transform these into the configuration space and add
 - Use the resulting force to move the robot (in its configuration space)

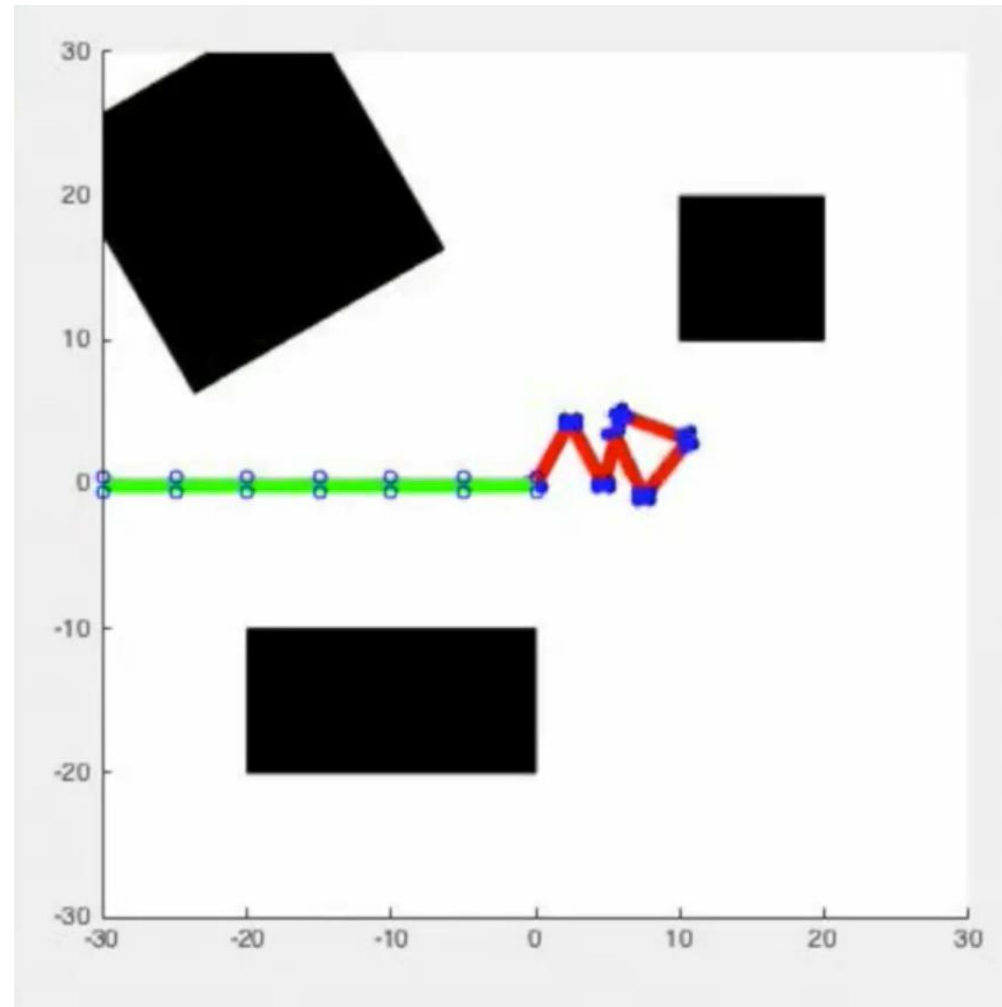




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Defining futures



Potential Fields

- **Potential Fields**

- Grid Based Approach
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- Potential Functions for Rigid Bodies
- **Summary**

Summary

- **Basic potential fields**
 - **attractive/repulsive forces**
- **Gradient following**
- **Navigation functions**
- **Extensions to more complex manipulators**



Defining futures